

Geophysical Research Letters®

RESEARCH LETTER

10.1029/2025GL118712

Key Points:

- Forced historical annual precipitation trends are broadly consistent across models, highlighting a robust atmospheric response to forcings
- Atmospheric internal variability shapes regional precipitation trend patterns, with a few ensemble members closely capturing observed trend
- Zonal-mean trends show “wet gets wetter, dry gets drier” paradigm across Northern Hemisphere land, and drying near 45°S linked to jet shift

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

W. Liang,
wengui.liang@lsu.edu

Citation:

Liang, W., Zhao, M., Tan, Z., Knutson, T., Dong, W., & Zhang, B. (2026). Confronting historical precipitation trends in models with observations: Forced signal and atmospheric internal variability. *Geophysical Research Letters*, 53, e2025GL118712. <https://doi.org/10.1029/2025GL118712>

Received 28 AUG 2025

Accepted 15 JAN 2026

© 2026. The Author(s).

This is an open access article under the terms of the [Creative Commons Attribution License](#), which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

Confronting Historical Precipitation Trends in Models With Observations: Forced Signal and Atmospheric Internal Variability

Wengui Liang^{1,2} , Ming Zhao³ , Zhihong Tan² , Thomas Knutson³ , Wenhao Dong⁴ , and Bosong Zhang² 

¹Department of Oceanography and Coastal Sciences, Louisiana State University, Baton Rouge, LA, USA, ²Program in Atmospheric and Oceanic Sciences, Princeton University, Princeton, NJ, USA, ³NOAA/Geophysical Fluid Dynamics Laboratory, Princeton, NJ, USA, ⁴Department of Earth System Science, Ministry of Education Key Laboratory for Earth System Modeling, Institute for Global Change Studies, Tsinghua University, Beijing, China

Abstract Future precipitation projections rely heavily on climate models, underscoring the need to evaluate their ability to simulate historical precipitation changes. Using multiple atmospheric models and ensemble simulations, we estimate the forced signals driven by sea surface warming and the direct effects of greenhouse gases and aerosols, as well as the atmospheric internal variability in precipitation trends since 1980. We find that forced precipitation trends are generally consistent across models, while atmospheric internal variability significantly influences regional patterns. Additionally, a few model members can reasonably well reproduce the observed pattern of precipitation trends. We highlight some regional wetting and drying are likely driven by forcings rather than the atmospheric internal variability. Zonal-mean trends over land reveal a “wet gets wetter, dry gets drier” paradigm in the Northern Hemisphere, while the Southern Hemisphere shows drying near 45°S associated with jet stream shifts. These results help improve our understanding of historical precipitation changes.

Plain Language Summary To improve our confidence in future precipitation changes, we need to ensure that climate models can reproduce observed historical precipitation trends. From an atmosphere-land-only perspective, historical trends reflect a combination of forced signals (e.g., greenhouse gases, aerosols, sea surface warming) and atmospheric internal variability (unforced natural and random shifts in wind, air pressure, and temperature patterns). We utilize multiple atmospheric models and their ensemble simulations to separate precipitation trends since 1980 into components driven by forcings and those due to atmospheric internal variability. We find that patterns of forced precipitation trends are similar across models, indicating a robust and strong forced signal. Atmospheric internal variability plays an important role in shaping the patterns of precipitation trends, and a few ensemble members are able to closely capture the observed trends. By comparing observed trends with forced trends across multiple models, we identify regional precipitation changes that are likely driven by forcings rather than internal variability. The zonal-mean precipitation trend over land follows a “wet-gets-wetter, dry-gets-drier” paradigm in the Northern Hemisphere, while in the Southern Hemisphere, the expected drying around 30°S shifts poleward to around 45°S, consistent with poleward shift of jet stream. These findings help us better understand historical precipitation changes.

1. Introduction

Numerous studies have explored the future changes in precipitation under a warming climate using climate models (e.g., Akinsanola et al., 2024; Grise, 2022; Liang & Zhang, 2022a; Pfahl et al., 2017). However, the accuracy of climate models in simulating historical precipitation trends remains uncertain, given the large discrepancies between model and observations (Shaw et al., 2024). The first-order precipitation changes in response to warming is an increase in the global mean, with regional patterns roughly following the “wet gets wetter, dry gets drier” (WWDD) paradigm, driven by enhanced capacity of the atmosphere to hold water vapor with relatively less change in atmospheric circulation (Held & Soden, 2006; Liang & Zhang, 2021, 2022b; Seager et al., 2014).

On a global scale, coupled ocean-atmosphere models can broadly reproduce the historical precipitation increase trends but tend to underestimate the magnitudes and have a drying bias over land compared to observations for the

past century (Knutson & Zeng, 2018; Y. Zhang et al., 2025). At regional scales, there are substantial discrepancies between observed and simulated precipitation trends (Makula & Zhou, 2022; Saha et al., 2014; Varuolo-Clarke et al., 2021), as historical precipitation changes driven by greenhouse gases (GHGs) could be obscured by the internal (natural) variability, uncertainty in aerosols, biases in model physics, and observation uncertainties (Masson-Delmotte et al., 2021; Shaw et al., 2024). Internal variability of the climate system, particularly on decadal to multidecadal timescales, could be large enough to dominate over regional precipitation trends induced by anthropogenic forcings during the 20th to 21st century (Dey et al., 2019; Huang et al., 2020; Varuolo-Clarke et al., 2021). The different impacts of various aerosols, such as black carbon and sulfate, pose more challenges for the models to accurately simulate the historical precipitation changes (Y. Zhang et al., 2025; S. Zhao & Suzuki, 2019; X. Zhao et al., 2020). In addition, observational uncertainties caused by historical changes in the spatial distribution and instrumentation of observation networks further complicate the assessment of the models' ability in simulating regional precipitation changes (Beck et al., 2017; Hegerl et al., 2015; Vose et al., 2014).

Overall, confronting models with observations and disentangling the roles of internal variability and anthropogenic forcings in historical precipitation changes remain challenging. To better assess the models' ability to simulate precipitation changes, this study employs several methodological approaches. Multiple climate models are utilized to reduce systematic model uncertainties and relatively recent observational data sets are incorporated to account for observational uncertainties. Furthermore, Atmospheric Model Intercomparison Project (AMIP) style prescribed sea surface temperature (SST) experiments are used to explicitly model the effect of SST-related internal variability and temporarily avoid the coupled model SST biases (Deser et al., 2012; Webb et al., 2017; M. Zhao & Knutson, 2024). These approaches allow us to evaluate how well climate models can reproduce observed precipitation change over recent decades when forced with observed SST variability, GHGs, aerosols, and other forcings, and to quantify the extent to which these changes are attributable to "external forcings" (including specified SSTs) versus atmospheric internal variability. The structure of the paper is as follows: Section 2 outlines the observations, reanalysis, climate models, and methodology. Section 3 presents the results. Section 4 provides a brief summary and discussion.

2. Data and Methods

To analyze historical precipitation trends and changes in atmospheric dynamic and thermodynamic properties, we utilized multiple observational and reanalysis data sets. For precipitation, we used four different data sets: (a) the Global Precipitation Climatology Project Version 2.3 (GPCP), providing monthly precipitation data starting from 1979 (Adler et al., 2018); (b) the Global Precipitation Climatology Centre (GPCC), which offers monthly land-only precipitation data beginning in 1891 (Schneider et al., 2014); (c) the Climate Prediction Center Merged Analysis of Precipitation (CMAP), with monthly precipitation available from 1979 (Xie & Arkin, 1997); and (d) the Multi-Source Weighted-Ensemble Precipitation (MSWEP) Version 2.8, which provides daily precipitation data starting from 1979 (Beck et al., 2019). For changes in circulation and atmospheric moisture content, we utilized three reanalysis data sets: the ECMWF Reanalysis Version 5 (ERA5), available from 1950 onward (Hersbach et al., 2020); the Modern-Era Retrospective Analysis for Research and Applications, Version 2 (MERRA-2), available from 1980 onward (Gelaro et al., 2017); and the Japanese 55-year Reanalysis (JRA-55), available from 1958 onward (Kobayashi et al., 2015).

To evaluate the models' ability in simulating precipitation changes, we utilized two sets of global atmospheric models with large ensemble AMIP simulations (see Table S1 in Supporting Information S1), covering the historical period from 1980 to 2014. These AMIP simulations are driven by observed monthly SSTs, sea ice concentrations, GHG concentrations, and aerosol emissions (Eyring et al., 2016; Liang et al., 2024; M. Zhao, 2022). The first set is produced by the high-resolution GFDL atmospheric model version 4.0 (C192AM4) with 17 ensemble members from 1980 to 2020 (M. Zhao, 2020; M. Zhao et al., 2018a). C192AM4 employs a cubed-sphere dynamical core with a horizontal resolution of approximately 50 km (M. Zhao, 2020, 2022). The second set consists of simulations from 1979 to 2014 from the latest AMIP6 archive (Webb et al., 2017), including 48 models and approximately 200 ensemble members with varying spatial resolutions (Table S1 in Supporting Information S1). AMIP6 is a component of the Coupled Model Intercomparison Project Phase 6 (CMIP6; Eyring et al., 2016). GFDL AM4.0 and AM4.1 are also part of AMIP6, they are essentially the same as C192AM4 except with a coarser (~100 km) horizontal resolution, and for AM4.1 it has a higher model top with more sophisticated atmospheric chemistry (Horowitz et al., 2020; M. Zhao et al., 2018b). For consistent comparison, all model simulations, observational data sets, and reanalysis are interpolated to a $1^\circ \times 1^\circ$ grid. Linear annual trends in

precipitation, precipitable water and 250 hPa zonal wind are calculated over the period 1980 to 2014. In addition, trends from 1980 to 2020 are also computed for GPCP and C192AM4 as a sensitivity test. Precipitation changes from 1980 to 2020 are largely consistent with those from 1980 to 2014 (Figure S1 in Supporting Information S1). In this study, we focus on the 1980–2014 period to include a larger number of models.

As SSTs are prescribed using observations in the AMIP simulations, they can be considered as an “external forcing” to the atmosphere. Therefore, in this study, we treat the atmosphere–land as a single system, and we consider the SST warming and the direct effects of GHGs and aerosols as forcings. In this framework, the internal variability is confined solely on atmospheric internal modes of variability, such as the Arctic Oscillation/North Atlantic Oscillation (AO/NAO; Ambaum et al., 2001), the Pacific–North American (PNA) pattern (W. Dong et al., 2024), and the Southern Annular Mode (SAM; Gillett et al., 2006); whereas the ocean–atmospheric coupled variability is considered as SST-forced atmospheric responses without atmospheric feedback on the ocean. For each model with multiple ensemble members, the forced precipitation trend driven by direct effect of CO₂ (Liang et al., 2024) and other GHGs, direct effect of aerosols (Y. Zhang et al., 2025), SST warming (M. Zhao, 2022), and other forcings is estimated using its ensemble mean, while the atmospheric internal variability is quantified as the deviation from the models ensemble mean (Deser et al., 2012). It is important to note that the total impacts of GHGs and aerosols include both the so-called “fast responses” and the long-term effect embedded in SST warming (L. Dong & Leung, 2021; Liang et al., 2024). That is, with prescribed SSTs, the non-SST-related impacts of GHGs and aerosols arise primarily through fast atmospheric responses, while slow responses are already implicitly included through imposed SST forcing (Richardson et al., 2016; Samset et al., 2016; S. Zhang et al., 2021). In this study we do not isolate individual forcings, but further work is needed to disentangle their contributions to historical precipitation trends.

To ensure a more robust estimation of forced precipitation trends, we only include ensemble means from models with at least nine ensemble members. Based on this criterion, eight models are selected for ensemble mean analysis. A sensitivity test was conducted using C192AM4 (17 members), GISS (20 members), and IPSL (22 members), in which nine members were randomly selected (Figure S2 in Supporting Information S1). The resulting mean trend patterns from these subsamples were found to be highly consistent, suggesting that nine ensemble members are sufficient to capture most of the forced signal.

3. Results

3.1. Forced Precipitation Trends

The patterns of forced annual precipitation trends from 1980 to 2014 are broadly consistent across eight different models, with pattern correlation coefficients between the multimodel mean and each individual model close to 0.9 (Figure 1). This suggests that the precipitation trends forced by GHGs, aerosols, SSTs, and other forcings are robust against atmospheric internal variability across models. The major features of the forced precipitation pattern in the tropics include a significant increase in the western Pacific, a decrease in the central Pacific, and increases over central America (Figure 1). This zonal pattern of precipitation changes may be associated with the La Nina-like SST warming pattern accompanied by a strengthening of the Walker Circulation in recent decades (Barichivich et al., 2018; Chadwick et al., 2014; Chung et al., 2019). In the extratropics, the forced trends show precipitation decreases over southern North America and the northeastern and southeastern Pacific Ocean, along with increases over the Sahel, Southern Ocean, and the Arctic. Regions with significant forced precipitation trends are defined as those where at least 80% of the ensemble members exhibit the same sign of trend as the ensemble mean (indicated by stippling in Figure 1). The high consistency among ensemble members suggests that the forced signal might play a dominant role over atmospheric internal variability. In these regions, assuming only a single ensemble member is available, the probability that it qualitatively represents the forced trend is greater than 80%. In contrast, for regions where internal variability dominates, multiple ensemble simulations are required to quantify the relative contributions of the forced signal and internal variability. Assuming the models realistically capture atmospheric internal variability, its dominance may suggest greater challenges for models in reproducing the observed precipitation trends.

Moderate similarity is found between the forced precipitation changes and observed precipitation trends (Figure 2). The pattern correlation coefficient of precipitation trends between the multi-model mean and GPCP is 0.592 (Figure 2), with some variation across individual models (Figure 1). This indicates that the forced signal in the precipitation changes over the recent decades is discernible despite atmospheric internal variability. In regions

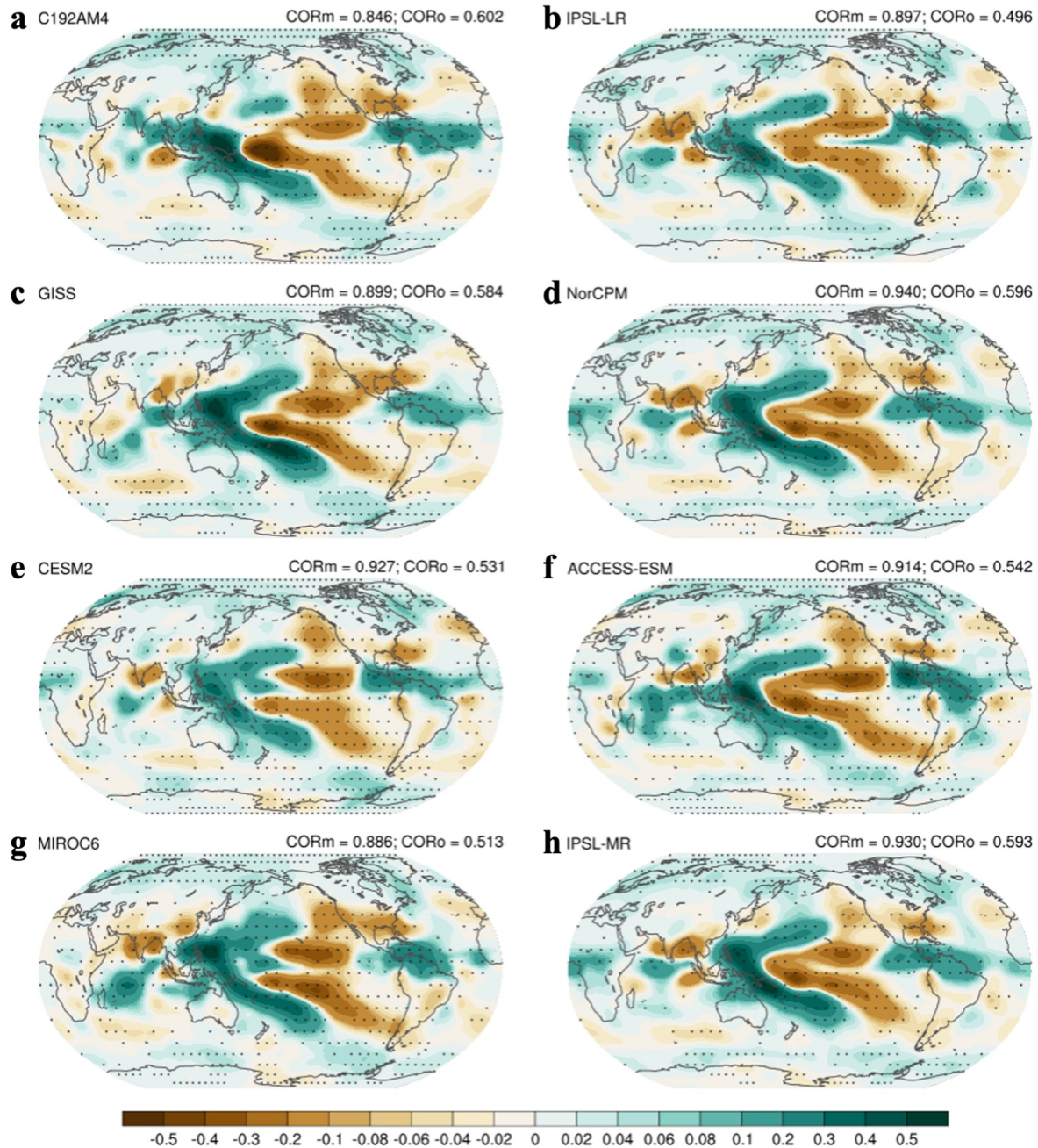


Figure 1. Ensemble-mean annual precipitation trends (mm/day/decade) from 1980 to 2014 for different AMIP6 models (Table S1 in Supporting Information S1). CORm denotes the pattern correlation coefficient between the ensemble mean of each AMIP6 model and the multi-model mean. CORo represents the global pattern correlation coefficient between the ensemble mean of each AMIP6 model and GPCP. Stippling indicates regions where at least 80% of ensemble members agree on the sign of the ensemble-mean trend.

where models show consistent forced signals across these models, we can reasonably assume that the models are capturing the forced component of historical precipitation changes, except if the majority of models have a common bias. It is worth noting that the spread among models in the forced precipitation trends is non-negligible, especially in the tropics, but the commonality remains significant relative to the inter-model differences in regions with significant forced trends (Figure 1 and Figure S3 in Supporting Information S1). The spatial patterns of percentage changes in precipitation remain largely consistent among different models, although the similarity between the model ensemble mean and GPCP is slightly lower than when examining the original changes (Figure S4 in Supporting Information S1).

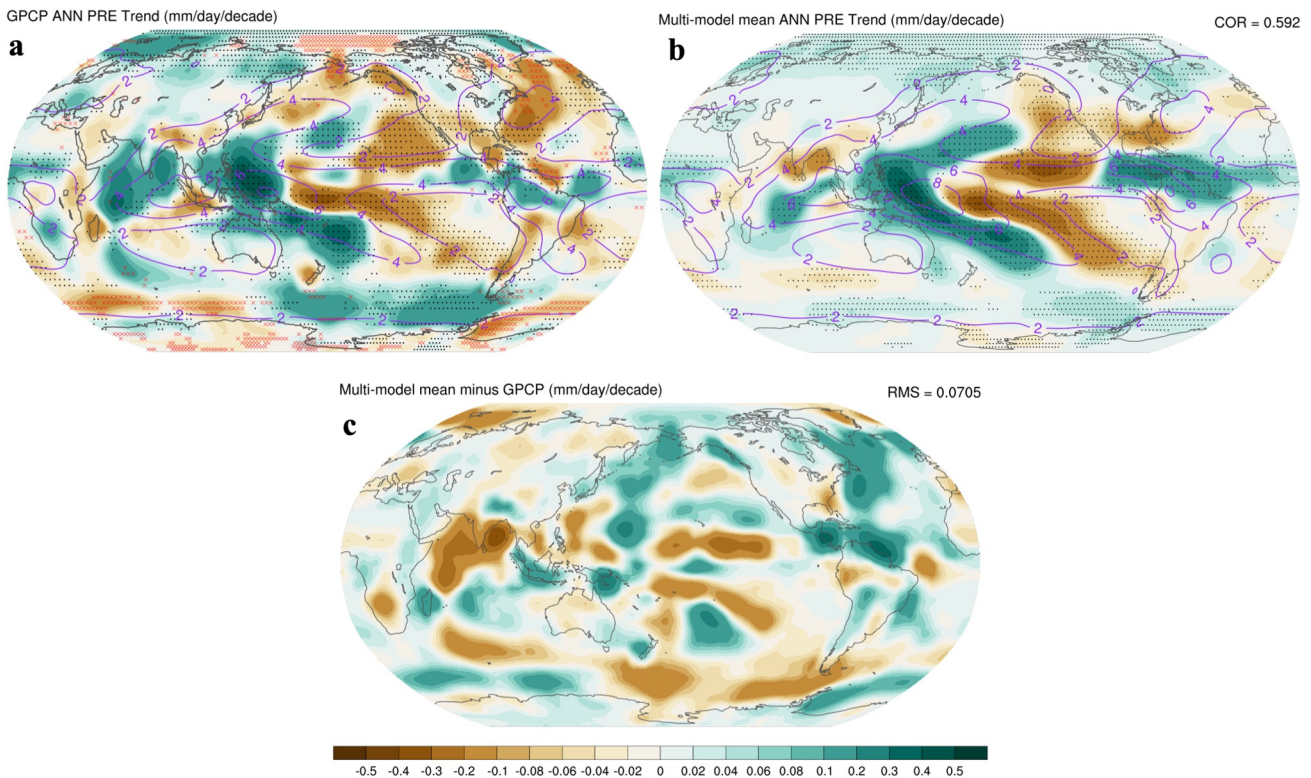


Figure 2. (a) Annual precipitation climatology (contours, mm/day) and trends (shading, mm/day/decade) from 1980 to 2014 for GPCP. (b) Same as (a), but for the AMIP6 multi-model mean. (c) Differences between multi-model mean in (b) and GPCP in (a). Black dots in (a) indicate regions where the GPCP precipitation trend and ensemble-mean trends of all eight models in Figure 1 share the same sign. Red crosses mark regions where the GPCP precipitation trend falls outside the range of AMIP6 models and is opposite to the multi-model mean trend. Black dots in (b) represent at least 80% of all ensemble members from AMIP6 agreeing on signs of changes. COR is the global pattern correlation coefficient between GPCP and the AMIP6 multi-model mean. RMS is the global root-mean-square difference between GPCP and the AMIP6 multi-model mean.

We can then assess the extent to which observed precipitation trends are driven by forcings rather than atmospheric internal variability by comparing observed changes (Figure 2a) to the modeled forced responses (Figures 1 and 2b). In the tropics, the strong similarity between observed and forced precipitation trends suggests that the observed changes are primarily driven by the La-Nina-like SST trend. Similarly, precipitation decreases over southern North America, southern South America, and the adjacent Pacific Ocean, as well as increases over northeastern Eurasia and the southern South Pacific, are likely attributable to forcings. Precipitation trends in other observational data sets that cover both land and ocean are generally consistent with those in GPCP, except for notable differences over the tropical oceans (Figure S5 in Supporting Information S1). It is important to note that in regions where the forced trends show significant differences among models, such as South Asia and the Indian Ocean (Figure 1), these discrepancies may result from differences in model physics or systematic biases in the simulations of precipitation trends, which require further investigation. Moreover, in some regions (mainly high latitudes), the observed trend falls outside the range of model ensemble variability, and the observed and modeled forced trends are in opposite directions (red stippling in Figure 2a). Given that confidence in high-latitude observations is relatively low, the discrepancies between the observed and modeled trends are likely dominated by observational uncertainties, but could also arise from model biases or issues in the observation–model comparison (Shaw et al., 2024).

3.2. Impacts of Atmospheric Internal Variability

Atmospheric internal variability plays an important role in determining the historical precipitation changes. Figure 2c shows there are significant differences between observed trends (Figure 2a) and modeled trends driven by observed SST and radiative forcings (Figure 2b). These discrepancies are common across different model members (Figure S6 in Supporting Information S1), which might be partially attributed to the atmospheric

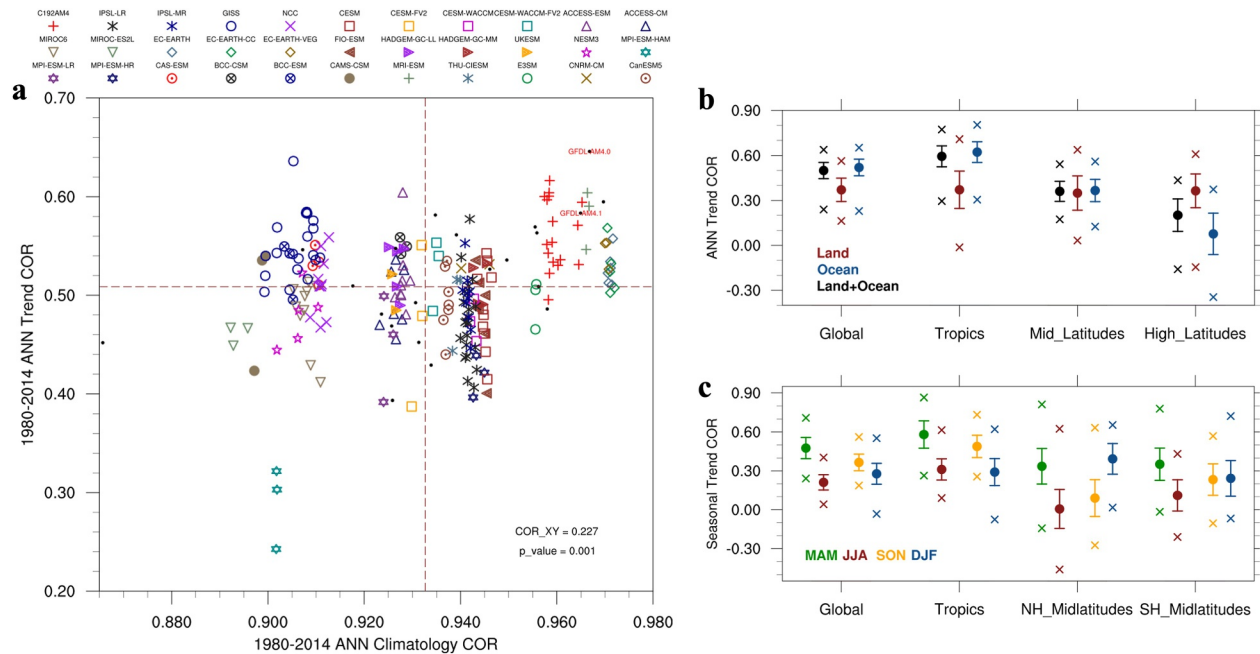


Figure 3. Scatter plot of pattern correlation coefficients between models and observation. (a) The Y-axis represents the pattern correlation coefficient between annual precipitation trends (1980–2014) from GPCP and all AMIP6 model ensemble members. The X-axis represents the same correlation coefficient but for climatological mean precipitation. To quantitatively demonstrate the influence of internal variability within the same model, models with more than three ensemble members are distinguished by different marker shapes and colors, while the remaining models are shown as black dots. GFDL-AM4.0 and GFDL-AM4.1 are highlighted in red text. Dashed red lines indicate the mean values along the X- and Y-axes. COR_{XY} and p -value denote the correlation coefficient and statistical significance between the X- and Y-axes, respectively. (b) Statistics of the pattern correlation coefficient between annual precipitation trends (1980–2014) from GPCP and all AMIP6 model ensemble members over land and ocean at different latitudes: tropics (lower than 23.5°), midlatitudes (between 23.5° and 60°), and high latitudes (higher than 60°). Solid dots represent the mean values, vertical bars indicate one standard deviation, and crosses mark the maximum and minimum values. (c) Same as (b), but showing seasonal statistics by latitudes. NH (SH) represents the Northern (Southern) Hemisphere. Seasons are defined as March–May (MAM), June–August (JJA), September–November (SON), and December–February (DJF).

internal variability during the observed historical period. They could also arise from other factors, including observational uncertainties (Beck et al., 2017; Hegerl et al., 2015) and the fact that models may not fully reproduce the forced signals or atmospheric internal variability present in the real world. Previous studies have shown that models often struggle to accurately represent aerosol forcings and their associated climate effects (Kok et al., 2023; Moseid et al., 2020; Wang et al., 2021; A. Zhao et al., 2022). To quantify the role of internal variability, we analyze individual ensemble members and check if some of them produce better pattern correlation with the observed precipitation trend than the ensemble mean. Figure 3a shows that the trend patterns vary significantly even among individual members of the same model, highlighting the impact of atmospheric internal variability, despite that the climatological mean precipitation pattern is broadly reproduced by all individual members. A few ensemble members from C192AM4, GFDL-AM4.0, GISS and ACCESS-ESM (Table S1 in Supporting Information S1) can capture the major features of the observed precipitation changes, with the pattern correlation coefficient with GPCP larger than 0.6 (Figure 3a). Which suggests that the atmospheric models possess some skills in simulating historical precipitation changes under realistic forcings from SST warming, aerosols, and GHGs, and that with sufficiently large ensemble simulations it is possible to reproduce the observed precipitation pattern. Similar results are obtained when using CMAP or MSWEP (Figure S7 in Supporting Information S1). It should be noted that while the regional pattern of future SST warming remains uncertain (M. Zhao & Knutson, 2024), which complicates the projections of future precipitation patterns. However, once the models are able to simulate the correct SST warming pattern, confidence in future precipitation projections can be improved.

The atmospheric internal variability has a greater influence on precipitation changes over land than over ocean, as indicated by the larger variance in pattern correlation coefficients with GPCP (Figure 3b, red). As the constraint by SST is likely stronger over the ocean, the modeled precipitation trends show better pattern correlations with GPCP, especially in the tropics (Figure 3b, blue). Such land–ocean contrasts in model performance for

precipitation trend simulations could also be linked to the stronger aerosol forcings over land than over ocean, given that climate models still face challenges in accurately representing aerosol–climate interactions. Previous studies have shown that aerosol emissions have substantial impacts on land precipitation over the past few decades (Y. Zhang et al., 2025; S. Zhao & Suzuki, 2019). Specifically, sulfate (SO_4) and black carbon have been identified as major contributors to the overestimated decline in land precipitation across East Asia, Africa, and western–central North America (Y. Zhang et al., 2025). Nonetheless, these studies also suggest that other forcing factors may play important roles in certain regions, and further research is needed to quantitatively attribute the land biases to the various forcing factors.

The models' ability to simulate observed precipitation tends to decrease with increasing latitude, consistent with the greater influence of atmospheric internal variability on decadal trends in the middle and high latitudes (Deser et al., 2012). The impacts of atmospheric internal variability on precipitation changes show little seasonality, as the variance of pattern correlation coefficients (vertical bars in Figure 3c) with GPCP are similar across seasons. Except at midlatitudes in both the Northern and Southern Hemispheres, atmospheric internal variability has a slightly greater impact on precipitation trends in summer than in winter, respectively. In contrast, model performance (solid dots in Figure 3c) in simulating historical precipitation changes varies by season. In the Northern Hemisphere midlatitudes, models perform better in boreal spring and winter than in boreal summer and fall. Compared to the Northern Hemisphere, model performance is better in the Southern Hemisphere midlatitudes in all seasons except austral winter.

3.3. The WWDD Paradigm and Poleward Shift of Jet Stream

Held and Soden (2006) proposed the simple WWDD paradigm based on zonal-mean precipitation changes, assuming minimal changes in atmospheric circulation. The WWDD paradigm is found to hold less robustly over land than over ocean, primarily due to changes in horizontal gradients of thermodynamic properties (Byrne & O'Gorman, 2015). For the relatively short time period from 1980 to 2014, we find that zonal-mean precipitation changes over land generally align with the WWDD paradigm: precipitation increases in the wet tropics and high latitudes, and decreases in the dry midlatitudes of the Northern Hemisphere (Figure 4a). These patterns are consistently seen in both observations and the model ensemble means, indicating that the observed WWDD-like changes are likely driven by forcings rather than atmospheric internal variability.

However, a notable deviation from the WWDD paradigm occurs in the Southern Hemisphere, where significant drying is observed in the climatologically wet region between 40°S and 50°S mainly over southern South America and New Zealand (Figure 2). This drying is primarily a forced signal, as it appears consistently in both observations and the model ensemble means. This result is consistent with Garreaud et al. (2020), who attributed the drying in central Chile to anthropogenic forcing and ocean-driven (natural) variability. Unlike the zonal-mean WWDD pattern driven by thermodynamics (Held & Soden, 2006), this regional drying is linked to dynamic changes (Figure 5). In particular, a significant poleward shift of the Southern Hemisphere jet stream (Figure 5a; Figure S8a in Supporting Information S1) likely reduces transient precipitation due to weakened eddy activity in regions where the jet weakens (Chang et al., 2012; Liang & Zhang, 2021). This jet shift has been attributed to both the Antarctic ozone hole (Grise et al., 2013; Li et al., 2021) and tropical warming (Woollings et al., 2023; Yang et al., 2020). Thermodynamically, the ensemble means of models show an increase in atmospheric moisture content over this region due to global warming, which should exert a positive effect on precipitation (Figure 5b; Figure S8b in Supporting Information S1). However, no significant change in moisture content is found in the reanalysis data sets. The discrepancy in moisture trends across different reanalysis products, as well as between reanalysis and models, warrants further investigation (Lou et al., 2025; Simpson et al., 2024). Moreover, both the WWDD-like precipitation pattern in the tropics and Northern Hemisphere and its poleward shift in the Southern Hemisphere persist on seasonal timescales (Figure S9 in Supporting Information S1), with the locations of precipitation increases and decreases migrating seasonally.

From a zonal-mean perspective, the historical precipitation increases over tropical land appear to be primarily driven by enhanced moisture flux convergence rather than changes in evaporation. In contrast, at middle and high latitudes, evaporation plays a more substantial role in driving precipitation changes over land (Figures 4b and 4c). Spatial patterns of modeled hydrological changes support this distinction: in large portions of the midlatitude land regions, evaporation changes contribute more than 50% to the total precipitation changes, whereas in tropical land regions, changes in total moisture convergence (divergence) are the dominant contributors (Figure S10 in

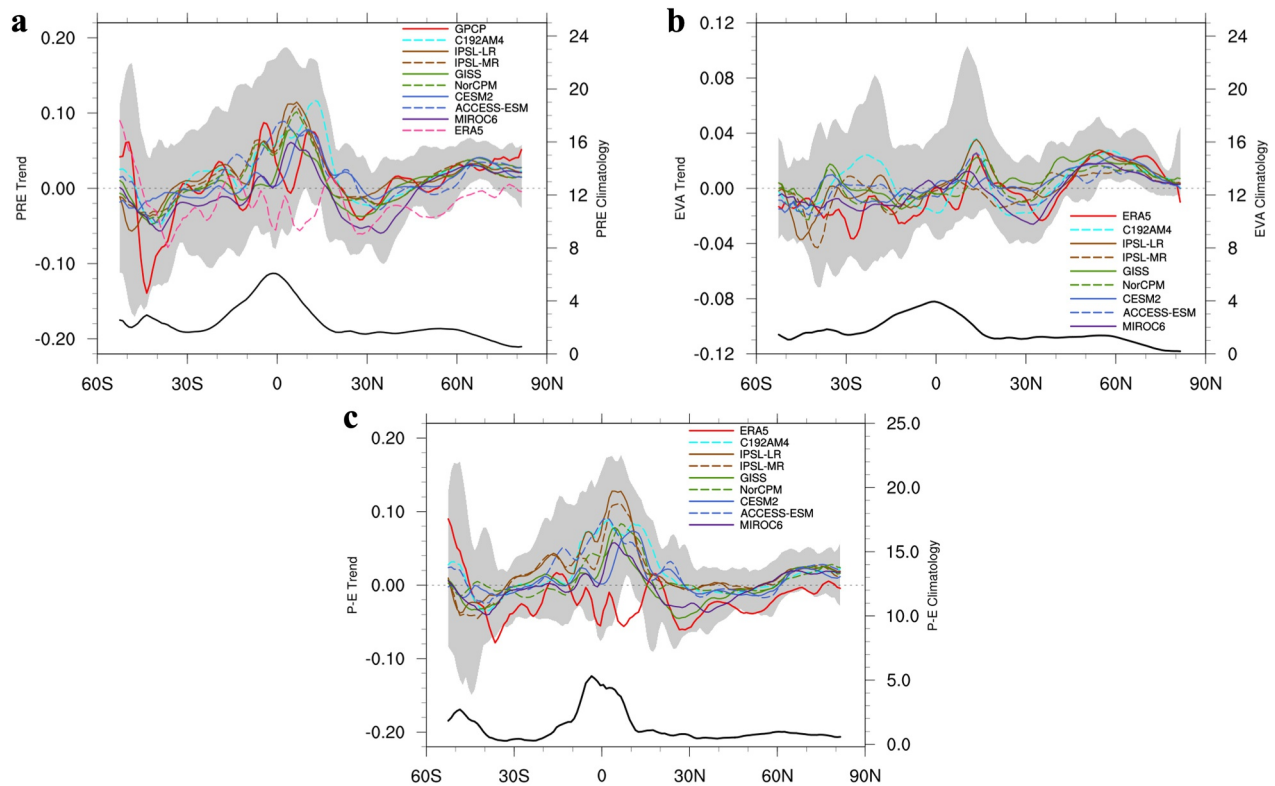


Figure 4. (a) Zonal-mean annual precipitation trends (mm/day/decade) over land from 1980 to 2014 based on GPCP, ERA5, and AMIP6 models (Table S1 in Supporting Information S1). Colored lines represent the ensemble means for models with more than nine ensemble members. The gray shading indicates the range of precipitation trends across all AMIP6 ensemble members. The black line shows the precipitation climatology (mm/day) from GPCP. The left y-axis is for trend and the right y-axis is for climatology. (b) Same as (a), but for evaporation trends (mm/day), with the black line representing evaporation climatology (mm/day) from ERA5. (c) Same as (a), but for trends in precipitation minus evaporation (P–E, mm/day). In (c), the black line represents P–E climatology (mm/day) from ERA5.

Supporting Information S1). For zonal-mean precipitation trends over the ocean (Figure S11 in Supporting Information S1), wetting trends are evident near the climatologically wettest regions around 5°N, whereas drying trends are observed around 5°S. Unlike over land regions (Figure 4a), the drying trends over the ocean around 30°N and 45°S simulated by the models are less pronounced. Overall, the discrepancies between models and observations or reanalyses are larger over the ocean than over land in the extratropics (Figure 4 and Figure S1 in Supporting Information S1). The observational uncertainty for precipitation is also larger over the ocean

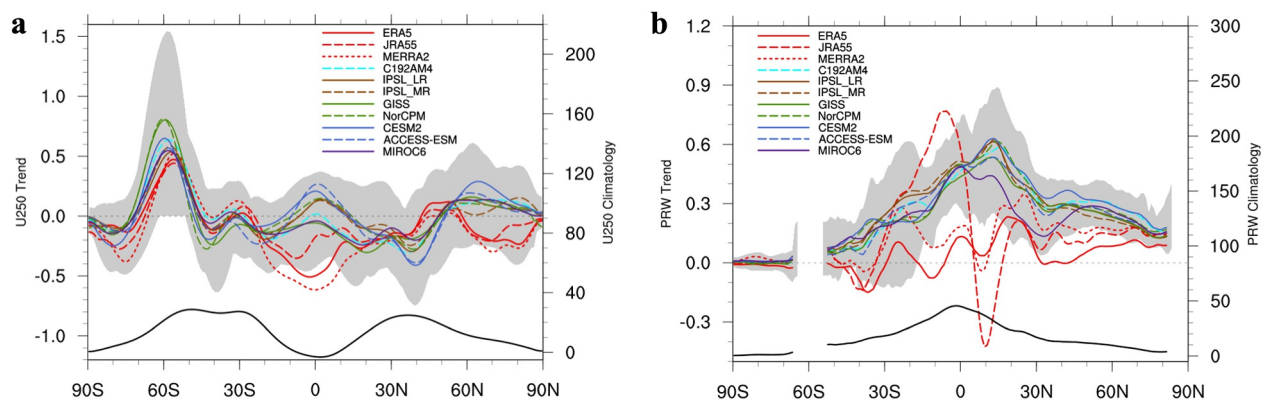


Figure 5. Same as Figure 4, but for annual trends in (a) 250 hPa zonal winds (U250, m/s/decade) and (b) precipitable water (PRW, kg/m²/decade). Trends from ERA5, JRA55, and MERRA2 are shown for comparison with AMIP6 models. Black lines represent climatology from ERA5. Trends in zonal wind are zonally averaged over both land and ocean while trends in precipitable water are over land only.

compared to land, as suggested by the spread across different data sets (Figures S12a and S12b in Supporting Information S1). In addition, evaporation estimates from different reanalysis products exhibit significant uncertainties over both land and ocean (Figures S12c and S12d in Supporting Information S1). These uncertainties partially contribute to the substantial discrepancies between reanalysis and the model ensemble means shown in Figures 4b and 4c, and Figures S11b, S11c in Supporting Information S1. For instance, trends in total moisture convergence in ERA5 differ substantially from those in model simulations (Figure 4c). In the tropics, models simulate a strong increase in total moisture convergence, whereas ERA5 shows a decrease. This discrepancy may stem from the global hydrological budget imbalance in ERA5 from 1980 to the mid-1990s, when precipitation was excessively reduced over land compared with GPCP and GPCC observations, while land evaporation showed little change (Hersbach et al., 2020). Therefore, caution is warranted when interpreting and comparing changes in evaporation and total moisture convergence between reanalysis and model simulations.

4. Summary and Discussion

We utilized multiple atmospheric models and their ensemble simulations, alongside observational data, to disentangle the forced signal (sum of GHGs, aerosols, SST warming, etc.) from atmospheric internal variability in historical precipitation trends from 1980 to 2014. We found that the forced precipitation trend patterns are broadly consistent across different atmospheric models, and some model members approximately capture the major pattern of observational precipitation trends. We further compared the forced trends in multiple models with the observations and identified regions experiencing drying or wetting trends that are likely attributable to forcings rather than atmospheric internal variability. Moreover, we found that atmospheric internal variability also heavily modulates the spatial patterns of precipitation trends, particularly over land regions.

From the zonal-mean perspective, we found that the climatologically wet tropics and high latitudes land regions in the Northern Hemisphere are becoming wetter, while the dry midlatitudes land regions in the Northern Hemisphere experience further drying since 1980. This is consistent with the WWDD theory, and it is primarily driven by forcings rather than atmospheric internal variability. However, we identify a notable shift in the Southern Hemisphere, characterized by drying around 45°S in climatologically wet regions, associated with the poleward displacement of the jet stream rather than moisture changes directly linked to warming. Moreover, the precipitation changes over land in the tropics appear to be attributed to changes in moisture flux convergence, while for the extratropics, the surface evaporation changes might play an important role in determining the hydroclimate changes. The aforementioned findings primarily focus on annual precipitation changes. However, precipitation exhibits strong seasonal cycles and regional variability (Grise, 2022; Liang et al., 2024; Liang & Zhang, 2021). Therefore, the results in this study are subject to seasonal and regional differences.

In this study, we utilize atmospheric models to investigate the impacts of atmospheric internal variability on relatively short-term precipitation changes over a few decades. It should be noted that SST-related internal variability such as the El Niño Southern Oscillation (ENSO), the Atlantic Multidecadal Oscillation (AMO), and the Pacific Decadal Oscillation (PDO) has significant influence on historical precipitation changes (Deser et al., 2012). Although we can use AMIP-style approach to avoid coupled model SST biases, SST-related variabilities are treated as forcing without interactive atmosphere-ocean coupled feedback. Further studies are needed to evaluate the ability of coupled ocean-atmosphere models to simulate historical precipitation trends, thereby enhancing our understanding of past changes and improving confidence in future precipitation projections.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

The AMIP6 and C192AM4 simulation data can be downloaded from Earth System Grid data portal: <https://esgf-node.llnl.gov/search/cmip6>. ERA5 reanalysis data is provided by Hersbach et al. (2023). MERRA2 reanalysis data is available from Global Modeling and Assimilation Office (GMAO) (2015). JRA55 reanalysis data can be downloaded at Japan Meteorological Agency (2013). GPCP is provided by NOAA PSL, Boulder, Colorado, USA, from their website at <https://psl.noaa.gov/data/gridded/data.gpcp.html>. GPCC is provided by Schneider

et al. (2011). CMAP observational data is provided by NOAA PSL, Boulder, Colorado, USA, from their website at <https://psl.noaa.gov/data/gridded/data.cmap.html>. MSWEP observational data is available at <https://www.glo-h2o.org/mswep/>. The GFDL AM4 model code is provided by Robinson et al. (2022).

Acknowledgments

We thank the anonymous reviewers for their valuable and constructive comments and suggestions. We thank Minghua Zhang, Brian Soden, Marc Prange and Xiuyuan Ding for helpful suggestions and comments. We would like to thank the AMIP modeling groups for generating and providing the model simulation data. We acknowledge the support from the leadership of NOAA OAR and GFDL for this project. Wengui Liang, Zhihong Tan, and Bosong Zhang were supported under award NA18OAR4320123 and NA23OAR4320198 from the National Oceanic and Atmospheric Administration, U.S. Department of Commerce. The statements, findings, conclusions, and recommendations are those of the authors and do not necessarily reflect the views of the National Oceanic and Atmospheric Administration, or the U.S. Department of Commerce.

References

- Adler, R. F., Sapiiano, M. R., Huffman, G. J., Wang, J. J., Gu, G., Bolvin, D., et al. (2018). The global precipitation climatology project (GPCP) monthly analysis (new version 2.3) and a review of 2017 global precipitation. *Atmosphere*, 9(4), 138. <https://doi.org/10.3390/atmos9040138>
- Akinsanola, A. A., Chen, Z., Kooperman, G. J., & Bobde, V. (2024). Robust future intensification of winter precipitation over the United States. *npj Climate and Atmospheric Science*, 7(1), 212. <https://doi.org/10.1038/s41612-024-00761-8>
- Ambaum, M. H., Hoskins, B. J., & Stephenson, D. B. (2001). Arctic oscillation or North Atlantic oscillation? *Journal of Climate*, 14(16), 3495–3507. [https://doi.org/10.1175/1520-0442\(2001\)014<3495:aonao>2.0.co;2](https://doi.org/10.1175/1520-0442(2001)014<3495:aonao>2.0.co;2)
- Barichivich, J., Gloor, E., Peylin, P., Brien, R. J., Schöngart, J., Espinoza, J. C., & Pattanyak, K. C. (2018). Recent intensification of Amazon flooding extremes driven by strengthened Walker circulation. *Science Advances*, 4(9), eaat8785. <https://doi.org/10.1126/sciadv.aat8785>
- Beck, H. E., Vergopolan, N., Pan, M., Levizzani, V., van Dijk, A. I. J. M., Weedon, G. P., et al. (2017). Global-scale evaluation of 22 precipitation datasets using gauge observations and hydrological modeling. *Hydrology and Earth System Sciences*, 21(12), 6201–6217. <https://doi.org/10.5194/hess-21-6201-2017>
- Beck, H. E., Pan, M., Roy, T., Weedon, G. P., Pappenberger, F., Van Dijk, A. I., et al. (2019). Daily evaluation of 26 precipitation datasets using Stage-IV gauge-radar data for the CONUS. *Hydrology and Earth System Sciences*, 23(1), 207–224. <https://doi.org/10.5194/hess-23-207-2019>
- Byrne, M. P., & O’Gorman, P. A. (2015). The response of precipitation minus evapotranspiration to climate warming: Why the “wet-get-wetter, dry-get-drier” scaling does not hold over land. *Journal of Climate*, 28(20), 8078–8092. <https://doi.org/10.1175/jcli-d-15-0369.1>
- Chadwick, R., Good, P., Andrews, T., & Martin, G. (2014). Surface warming patterns drive tropical rainfall pattern responses to CO₂ forcing on all timescales. *Geophysical Research Letters*, 41(2), 610–615. <https://doi.org/10.1002/2013gl058504>
- Chang, E. K., Guo, Y., & Xia, X. (2012). CMIP5 multimodel ensemble projection of storm track change under global warming. *Journal of Geophysical Research*, 117(D23). <https://doi.org/10.1029/2012jd018578>
- Chung, E. S., Timmermann, A., Soden, B. J., Ha, K. J., Shi, L., & John, V. O. (2019). Reconciling opposing Walker circulation trends in observations and model projections. *Nature Climate Change*, 9(5), 405–412. <https://doi.org/10.1038/s41558-019-0446-4>
- Deser, C., Phillips, A., Bourdette, V., & Teng, H. (2012). Uncertainty in climate change projections: The role of internal variability. *Climate Dynamics*, 38(3–4), 527–546. <https://doi.org/10.1007/s00382-010-0977-x>
- Dey, R., Lewis, S. C., Arblaster, J. M., & Abram, N. J. (2019). A review of past and projected changes in Australia’s rainfall. *WIREs Climate Change*, 10(3), 577. <https://doi.org/10.1002/wcc.577>
- Dong, L., & Leung, L. R. (2021). Winter precipitation changes in California under global warming: Contributions of CO₂, uniform SST warming, and SST change patterns. *Geophysical Research Letters*, 48(5), e2020GL091736. <https://doi.org/10.1029/2020gl091736>
- Dong, W., Zhao, M., Tan, Z., & Ramaswamy, V. (2024). Atmospheric rivers over eastern US affected by Pacific/North America pattern. *Science Advances*, 10(4), ead3325. <https://doi.org/10.1126/sciadv.adj3325>
- Eyring, V., Bony, S., Meehl, G. A., Senior, C. A., Stevens, B., Stouffer, R. J., & Taylor, K. E. (2016). Overview of the coupled model inter-comparison project phase 6 (CMIP6) experimental design and organization. *Geoscientific Model Development*, 9(5), 1937–1958. <https://doi.org/10.5194/gmd-9-1937-2016>
- Garreaud, R. D., Boisier, J. P., Rondanelli, R., Montecinos, A., Sepúlveda, H. H., & Veloso-Aguila, D. (2020). The central Chile mega drought (2010–2018): A climate dynamics perspective. *International Journal of Climatology*, 40(1), 421–439. <https://doi.org/10.1002/joc.6219>
- Gelaro, R., McCarty, W., Suárez, M. J., Todling, R., Molod, A., Takacs, L., et al. (2017). The modern-era retrospective analysis for research and applications, version 2 (MERRA-2). *Journal of Climate*, 30(14), 5419–5454. <https://doi.org/10.1175/jcli-d-16-0758.1>
- Gillett, N. P., Kell, T. D., & Jones, P. D. (2006). Regional climate impacts of the Southern Annular Mode. *Geophysical Research Letters*, 33(23). <https://doi.org/10.1029/2006gl027721>
- Global Modeling and Assimilation Office (GMAO). (2015). MERRA-2 instM_3d_asm_Np: 3d,Monthly mean,Instantaneous,Pressure-Level, Assimilation,Assimilated Meteorological Fields V5.12.4 [Dataset]. *Greenbelt, MD, USA, Goddard Earth Sciences Data and Information Services Center (GES DISC)*. <https://doi.org/10.5067/2E096JV59PK7>
- Grise, K. M. (2022). Atmospheric circulation constraints on 21st century seasonal precipitation storylines for the southwestern United States. *Geophysical Research Letters*, 49(17), e2022GL099443. <https://doi.org/10.1029/2022gl099443>
- Grise, K. M., Polvani, L. M., Tselioudis, G., Wu, Y., & Zelinka, M. D. (2013). The ozone hole indirect effect: Cloud-radiative anomalies accompanying the poleward shift of the eddy-driven jet in the Southern Hemisphere. *Geophysical Research Letters*, 40(14), 3688–3692. <https://doi.org/10.1002/grl.50675>
- Hegerl, G. C., Black, E., Allan, R. P., Ingram, W. J., Polson, D., Trenberth, K. E., et al. (2015). Challenges in quantifying changes in the global water cycle. *Bulletin of the American Meteorological Society*, 96(7), 1097–1115. <https://doi.org/10.1175/bams-d-13-00212.1>
- Held, I. M., & Soden, B. J. (2006). Robust responses of the hydrological cycle to global warming. *Journal of Climate*, 19(21), 5686–5699. <https://doi.org/10.1175/jcli3990.1>
- Hersbach, H., Bell, B., Berrisford, P., Biavati, G., Horányi, A., Muñoz Sabater, J., et al. (2023). ERA5 monthly averaged data on pressure levels from 1940 to present [Dataset]. *Copernicus Climate Change Service (C3S) Climate Data Store (CDS)*. <https://doi.org/10.24381/cds.6860a573>
- Hersbach, H., Bell, B., Berrisford, P., Hirahara, S., Horányi, A., Muñoz-Sabater, J., et al. (2020). The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730), 1999–2049. <https://doi.org/10.1002/qj.3803>
- Horowitz, L. W., Naik, V., Paulot, F., Ginoux, P. A., Dunne, J. P., Mao, J., et al. (2020). The GFDL global atmospheric chemistry-climate model AM4. 1: Model description and simulation characteristics. *Journal of Advances in Modeling Earth Systems*, 12(10), e2019MS002032. <https://doi.org/10.1029/2019ms002032>
- Huang, X., Zhou, T., Turner, A., Dai, A., Chen, X., Clark, R., et al. (2020). The recent decline and recovery of Indian summer monsoon rainfall: Relative roles of external forcing and internal variability. *Journal of Climate*, 33(12), 5035–5060. <https://doi.org/10.1175/JCLI-D-19-0833.1>
- Japan Meteorological Agency. (2013). JRA-55: Japanese 55-year reanalysis, monthly means and variances [Dataset]. *NSF National Center for Atmospheric Research*. <https://doi.org/10.5065/D60G3H5B>
- Knutson, T. R., & Zeng, F. (2018). Model assessment of observed precipitation trends over land regions: Detectable human influences and possible low bias in model trends. *Journal of Climate*, 31(12), 4617–4637. <https://doi.org/10.1175/jcli-d-17-0672.1>

- Kobayashi, S., Ota, Y., Harada, Y., Ebata, A., Moriya, M., Onoda, H., et al. (2015). The JRA-55 reanalysis: General specifications and basic characteristics. *Journal of the Meteorological Society of Japan. Series II*, 93(1), 5–48. <https://doi.org/10.2151/jmsj.2015-001>
- Kok, J. F., Storelvmo, T., Karydis, V. A., Adebisi, A. A., Mahowald, N. M., Evan, A. T., et al. (2023). Mineral dust aerosol impacts on global climate and climate change. *Nature Reviews Earth & Environment*, 4(2), 71–86. <https://doi.org/10.1038/s43017-022-00379-5>
- Li, S., Liu, W., Lyu, K., & Zhang, X. (2021). The effects of historical ozone changes on Southern Ocean heat uptake and storage. *Climate Dynamics*, 57(7), 2269–2285. <https://doi.org/10.1007/s00382-021-05803-y>
- Liang, W., & Zhang, M. (2021). Summer and winter precipitation in East Asia scale with global warming at different rates. *Communications Earth & Environment*, 2(1), 150. <https://doi.org/10.1038/s43247-021-00219-2>
- Liang, W., & Zhang, M. (2022a). Future changes of the eddy moisture convergence in winter over coastal lands in Eastern North America and East Asia. *Journal of Geophysical Research: Atmospheres*, 127(17), e2022JD037149. <https://doi.org/10.1029/2022jd037149>
- Liang, W., & Zhang, M. (2022b). Transient precipitation increase during winter in the Eastern North America. *Geophysical Research Letters*, 49(6), e2022GL098188. <https://doi.org/10.1029/2022gl098188>
- Liang, W., Zhao, M., Tan, Z., Knutson, T., Dong, W., & Zhang, B. (2024). The direct radiative effect of CO₂ increase on summer precipitation in North America. *Geophysical Research Letters*, 51(14), e2024GL109202. <https://doi.org/10.1029/2024gl109202>
- Lou, J., Joh, Y., & Delworth, T. L. (2025). Identifying common precursors of decadal vapor pressure deficit variability in the Southwestern United States. *Journal of Climate*.
- Makula, E. K., & Zhou, B. (2022). Coupled model intercomparison project phase 6 evaluation and projection of east African precipitation. *International Journal of Climatology*, 42(4), 2398–2412. <https://doi.org/10.1002/joc.7373>
- Masson-Delmotte, V., Zhai, P., Pirani, A., Connors, S. L., Péan, C., Berger, S., et al. (2021). Climate change 2021: The physical science basis. Contribution of working group I to the sixth assessment report of the intergovernmental panel on climate change (Vol. 2, p. 2391).
- Moseid, K. O., Schulz, M., Storelvmo, T., Julsrud, I. R., Olivie, D., Nabat, P., et al. (2020). Bias in CMIP6 models compared to observed regional dimming and brightening trends (1961–2014). *Atmospheric Chemistry and Physics Discussions*, 2020, 1–20.
- Pfahl, S., O’Gorman, P. A., & Fischer, E. M. (2017). Understanding the regional pattern of projected future changes in extreme precipitation. *Nature Climate Change*, 7(6), 423–427. <https://doi.org/10.1038/nclimate3287>
- Richardson, T. B., Forster, P. M., Andrews, T., & Parker, D. J. (2016). Understanding the rapid precipitation response to CO₂ and aerosol forcing on a regional scale. *Journal of Climate*, 29(2), 583–594. <https://doi.org/10.1175/jcli-d-15-0174.1>
- Robinson, T., Radhakrishnan, A., & Seth, U. (2022). NOAA-GFDL/AM4: Non-hydrostatic aquaplanet MG (highres_aquaplanet_2022) [Software]. *Zenodo*. <https://doi.org/10.5281/zenodo.7476908>
- Saha, A., Ghosh, S., Sahana, A. S., & Rao, E. P. (2014). Failure of CMIP5 climate models in simulating post-1950 decreasing trend of Indian monsoon. *Geophysical Research Letters*, 41(20), 7323–7330. <https://doi.org/10.1002/2014gl061573>
- Samset, B. H., Myhre, G., Forster, P. M., Hodnebrog, Ø., Andrews, T., Faluvegi, G., et al. (2016). Fast and slow precipitation responses to individual climate forcings: A PDRMIP multimodel study. *Geophysical Research Letters*, 43(6), 2782–2791. <https://doi.org/10.1002/2016gl068064>
- Schneider, U., Becker, A., Finger, P., Meyer-Christoffer, A., Rudolf, B., & Ziese, M. (2011). GPCC Full Data Reanalysis Version 6.0 at 1.0°: Monthly Land-Surface Precipitation from Rain-Gauges built on GTS-based and Historic Data [Dataset]. https://doi.org/10.5676/DWD_GPCC/FD_M_V7_100
- Schneider, U., Becker, A., Finger, P., Meyer-Christoffer, A., Ziese, M., & Rudolf, B. (2014). GPCC’s new land surface precipitation climatology based on quality-controlled in situ data and its role in quantifying the global water cycle. *Theoretical and Applied Climatology*, 115(1–2), 15–40. <https://doi.org/10.1007/s00704-013-0860-x>
- Seager, R., Neelin, D., Simpson, I., Liu, H., Henderson, N., Shaw, T., et al. (2014). Dynamical and thermodynamical causes of large-scale changes in the hydrological cycle over North America in response to global warming. *Journal of Climate*, 27(20), 7921–7948. <https://doi.org/10.1175/jcli-d-14-00153.1>
- Shaw, T. A., Arias, P. A., Collins, M., Coumou, D., Diedhiou, A., Garfinkel, C. I., et al. (2024). Regional climate change: Consensus, discrepancies, and ways forward. *Frontiers in climate*, 6, 1391634. <https://doi.org/10.3389/fclim.2024.1391634>
- Simpson, I. R., McKinnon, K. A., Kennedy, D., Lawrence, D. M., Lehner, F., & Seager, R. (2024). Observed humidity trends in dry regions contradict climate models. *Proceedings of the National Academy of Sciences*, 121(1), e2302480120. <https://doi.org/10.1073/pnas.2302480120>
- Varuolo-Clarke, A. M., Smerdon, J. E., Williams, A. P., & Seager, R. (2021). Gross discrepancies between observed and simulated twentieth-to-twenty-first-century precipitation trends in southeastern South America. *Journal of Climate*, 34(15), 6441–6457. <https://doi.org/10.1175/jcli-d-20-0746.1>
- Vose, R. S., Applequist, S., Squires, M., Durre, I., Menne, M. J., Williams, C. N., Jr., et al. (2014). Improved historical temperature and precipitation time series for U.S. climate divisions. *Journal of Applied Meteorology and Climatology*, 53(5), 1232–1251. <https://doi.org/10.1175/JAMC-D-13-0248.1>
- Wang, Z., Lin, L., Xu, Y., Che, H., Zhang, X., Zhang, H., et al. (2021). Incorrect Asian aerosols affecting the attribution and projection of regional climate change in CMIP6 models. *npj Climate and Atmospheric Science*, 4(1), 2. <https://doi.org/10.1038/s41612-020-00159-2>
- Webb, M. J., Andrews, T., Bodas-Salcedo, A., Bony, S., Bretherton, C. S., Chadwick, R., et al. (2017). The cloud feedback model intercomparison project (CFMIP) contribution to CMIP6. *Geoscientific Model Development*, 10(1), 359–384. <https://doi.org/10.5194/gmd-10-359-2017>
- Woollings, T., Drouard, M., O’Reilly, C. H., Sexton, D. M., & McSweeney, C. (2023). Trends in the atmospheric jet streams are emerging in observations and could be linked to tropical warming. *Communications Earth & Environment*, 4(1), 125. <https://doi.org/10.1038/s43247-023-00792-8>
- Xie, P., & Arkin, P. A. (1997). Global precipitation: A 17-year monthly analysis based on gauge observations, satellite estimates, and numerical model outputs. *Bulletin of the American Meteorological Society*, 78(11), 2539–2558. [https://doi.org/10.1175/1520-0477\(1997\)078<2539:gpayma>2.0.co;2](https://doi.org/10.1175/1520-0477(1997)078<2539:gpayma>2.0.co;2)
- Yang, D., Arblaster, J. M., Meehl, G. A., England, M. H., Lim, E. P., Bates, S., & Rosenbloom, N. (2020). Role of tropical variability in driving decadal shifts in the Southern Hemisphere summertime eddy-driven jet. *Journal of Climate*, 33(13), 5445–5463. <https://doi.org/10.1175/jcli-d-19-0604.1>
- Zhang, S., Stier, P., & Watson-Parris, D. (2021). On the contribution of fast and slow responses to precipitation changes caused by aerosol perturbations. *Atmospheric Chemistry and Physics*, 21(13), 10179–10197. <https://doi.org/10.5194/acp-21-10179-2021>
- Zhang, Y., Knutson, T. R., Shevliakova, E., Horowitz, L. W., Ginoux, P., Paynter, D., & Westervelt, D. M. (2025). The fast response of land precipitation to historical anthropogenic black carbon and sulfate aerosols in the GFDL ESM4 climate model. *Journal of Climate*, 38(10), 2395–2411. <https://doi.org/10.1175/jcli-d-24-0240.1>
- Zhao, A., Ryder, C. L., & Wilcox, L. J. (2022). How well do the CMIP6 models simulate dust aerosols? *Atmospheric Chemistry and Physics*, 22(3), 2095–2119. <https://doi.org/10.5194/acp-22-2095-2022>

- Zhao, M. (2020). Simulations of atmospheric rivers, their variability, and response to global warming using GFDL's new high-resolution general circulation model. *Journal of Climate*, 33(23), 10287–10303. <https://doi.org/10.1175/JCLI-D-20-0241.1>
- Zhao, M. (2022). A study of AR-TS-and MCS-associated precipitation and extreme precipitation in present and warmer climates. *Journal of Climate*, 35(2), 479–497. <https://doi.org/10.1175/jcli-d-21-0145.1>
- Zhao, M., Golaz, J. C., Held, I. M., Guo, H., Balaji, V., Benson, R., et al. (2018a). The GFDL global atmosphere and land model AM4. 0/LM4. 0: 1. Simulation characteristics with prescribed SSTs. *Journal of Advances in Modeling Earth Systems*, 10(3), 691–734. <https://doi.org/10.1002/2017MS001208>
- Zhao, M., Golaz, J. C., Held, I. M., Guo, H., Balaji, V., Benson, R., et al. (2018b). The GFDL global atmosphere and land model AM4. 0/LM4. 0: 2. Model description, sensitivity studies, and tuning strategies. *Journal of Advances in Modeling Earth Systems*, 10(3), 735–769. <https://doi.org/10.1002/2017MS001209>
- Zhao, M., & Knutson, T. (2024). Crucial role of sea surface temperature warming patterns in near-term high-impact weather and climate projection. *npj Climate and Atmospheric Science*, 7(1), 130. <https://doi.org/10.1038/s41612-024-00681-7>
- Zhao, S., & Suzuki, K. (2019). Differing impacts of black carbon and sulfate aerosols on global precipitation and the ITCZ location via atmosphere and ocean energy perturbations. *Journal of Climate*, 32(17), 5567–5582. <https://doi.org/10.1175/jcli-d-18-0616.1>
- Zhao, X., Allen, R. J., Wood, T., & Maycock, A. C. (2020). Tropical belt width proportionately more sensitive to aerosols than Greenhouse Gases. *Geophysical Research Letters*, 47(7), e2019GL086425. <https://doi.org/10.1029/2019gl086425>